PCCST501 - Computer Networks

Assignment 3: Data-Link Layer, Ethernet and Wireless LANs

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Q1. Data-Link Layer Duties; LLC and MAC

The primary responsibility of the Data-Link Layer is the node-to-node delivery of data across a direct physical connection. It acts as an intermediary, taking packets from the Network Layer and encapsulating them into frames for physical transmission, shielding higher layers from the complexities of raw bit signaling. Its core duties include:

- **Framing:** Adding headers and trailers to data payloads to demarcate the start and end of a transmission block.
- **Physical Addressing:** Assigning source and destination MAC addresses to uniquely identify the physical hardware communicating on the link.
- **Error Detection:** Using mechanisms like the CRC-32 checksum to identify and discard frames that become corrupted during transit.
- **Flow Control:** Ensuring a high-speed transmitter does not overwhelm a slower receiver with too much data.
- **Medium Access Control (MAC):** Determining how devices share a common communication medium to prevent data collisions.

To efficiently manage these tasks, the layer is divided into two specialized sub-layers:

Feature	LLC (Logical Link Control - 802.2)	MAC (Media Access Control)
Position	Upper sub-layer, interfacing with the Network Layer.	Lower sub-layer, interfacing with the Physical Layer.
Primary Role	Identifies the network protocol being carried (via SAP/SNAP) and can handle flow control.	Constructs frames, appends physical addresses, calculates FCS, and manages medium access.
Medium Dependency	Independent; operates uniformly across all LAN types.	Dependent; varies significantly between wired (802.3) and wireless (802.11).
Examples	Specifying whether the payload is an IPv4 or ARP packet.	CSMA/CD (traditional Ethernet) or CSMA/CA (Wi-Fi).

Q2. Ethernet Frame Format

An Ethernet II frame consists of several specific fields designed to synchronize communication, identify endpoints, and verify data integrity. The total size of a standard frame (excluding the preamble and SFD) ranges from a minimum of 64 bytes to a maximum of 1518 bytes.

Field (Size)	Description & Purpose
Preamble (7 B)	Alternating binary 1s and 0s used to synchronize

	receiver clocks.
SFD (1 B)	Start Frame Delimiter (10101011); indicates the actual frame is beginning.
Destination MAC (6 B)	Target hardware address (can be unicast, multicast, or broadcast FF:FF:FF:FF:FF:FF).
Source MAC (6 B)	Sender's hardware address; used by switches to populate their MAC tables.
EtherType (2 B)	Identifies the encapsulated protocol (e.g., 0x0800 for IPv4, 0x0806 for ARP).
Payload (46–1500 B)	The actual data. Must be padded to at least 46 bytes; standard max is 1500 bytes.
FCS (4 B)	Frame Check Sequence (CRC-32) covering the frame to detect transmission errors.

Q3. Wireshark Capture and MAC Addresses Analysis

Methodology: Wireshark was initiated on an active network interface. Network traffic was intentionally generated using commands like ping 8.8.8.8 and arp -a, along with general web requests. After capturing for roughly 30 seconds, the filter arp || icmp || dns was applied to isolate relevant packets.

Note: The following data represents a documented sample analysis of five specific network frames observed during such a capture.

Frame #	Protocol	Source MAC	Destination MAC	EtherType
1	ARP Request	aa:bb:cc:11:22:33	ff:ff:ff:ff:ff:ff	0x0806
2	ARP Reply	00:11:22:44:55:66	aa:bb:cc:11:22:33	0x0806
3	ICMP Request	aa:bb:cc:11:22:33	00:11:22:44:55:66	0x0800
4	ICMP Reply	00:11:22:44:55:66	aa:bb:cc:11:22:33	0x0800
5	DNS Query	aa:bb:cc:11:22:33	00:11:22:44:55:66	0x0800

Observations:

- The initial ARP request utilizes the broadcast MAC address (ff:ff:ff:ff:ff:ff) because the sender does not yet know the gateway's hardware address. The subsequent reply is unicast directly back to the sender.
- During the ICMP request (ping) to an external IP (8.8.8.8), the destination MAC address belongs to the local gateway router, not the Google server. This demonstrates that MAC addresses are only relevant for local-link delivery, while IP addresses handle end-to-end routing.
- The first three octets of any given MAC address represent the Organizationally Unique Identifier (OUI), which identifies the manufacturer of the network interface card.

Q4. Switch MAC Address Learning Mechanism

A switch dynamically builds a MAC address table (or CAM table) that associates specific hardware addresses with its physical ports. It processes incoming frames using four primary rules:

1. Learn: The switch records the source MAC address and binds it to the port where the frame arrived.
2. Forward: If the destination MAC is already in the table, it sends the frame exclusively out of that specific port.

3. Flood: If the destination MAC is unknown, or if it is a broadcast/multicast frame, it sends the frame out of all ports except the arrival port.
4. Filter: If the destination is known to be on the same port the frame arrived on, the switch drops it.

Step-by-Step Scenario (Hosts A, B, C, D on Ports 1, 2, 3, 4):

- **A sends to B:** Switch learns A on Port 1. B is unknown, so it floods the frame to ports 2, 3, and 4. (Table: A→1)
- **B replies to A:** Switch learns B on Port 2. A is known, so it forwards solely to Port 1. (Table: A→1, B→2)
- **C sends to A:** Switch learns C on Port 3. A is known, so it forwards solely to Port 1. (Table: A→1, B→2, C→3)
- **D sends to B:** Switch learns D on Port 4. B is known, so it forwards solely to Port 2. (Table: A→1, B→2, C→3, D→4)

Q5. Comparing Hubs, Bridges, and Switches

Attribute	Hub	Bridge	Switch
OSI Layer	Layer 1 (Physical)	Layer 2 (Data Link)	Layer 2 (Data Link) / Layer 3
Forwarding	Repeats electrical signals to all connected ports.	Uses a software MAC table to filter traffic between segments.	Uses ASICs (hardware) to switch traffic intelligently per port.
Collision Domains	One shared domain.	One per segment.	One independent domain per port.
Broadcast Domains	One shared domain.	One shared domain.	One per VLAN.
Bandwidth	Shared across all active users.	Shared per physical segment.	Dedicated bandwidth for every port.
Relevance	Completely obsolete.	Replaced by modern switches.	The standard foundation of modern LANs.

Q6. The Decline of CSMA/CD in Modern Networks

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) was the access protocol for legacy, half-duplex Ethernet over shared media (like coaxial cables or hubs). Devices would listen to the line; if quiet, they transmitted. If two devices transmitted simultaneously, a collision occurred. Both would send a “jam signal” and pause for a random duration (using binary exponential back-off) before attempting retransmission.

Why it is largely obsolete today:

- **Micro-segmentation:** Modern networks use switches instead of hubs. A switch dedicates a separate collision domain to every single port, isolating devices from one another.
- **Full-Duplex Communication:** Modern connections utilize dedicated pairs of wires for transmitting and receiving simultaneously. Since data flows in both directions without interference, physical collisions are impossible, rendering the “Collision Detection” aspect unnecessary.
- **Speed Requirements:** Gigabit Ethernet and faster standards mandate full-duplex operation, inherently bypassing the need for CSMA/CD.

Q7. CSMA/CA in 802.11 vs. CSMA/CD

Wireless environments cannot reliably detect collisions while transmitting because the radio cannot listen and transmit simultaneously on the same frequency. Furthermore, the “hidden node problem” means two devices

might both reach an access point without hearing each other. Therefore, Wi-Fi uses CSMA/CA (Collision Avoidance) via the Distributed Coordination Function (DCF):

5. The device senses the channel physically and virtually (via the Network Allocation Vector, NAV).
6. If idle, it waits a standard DIFS period. If busy, it waits until the channel clears.
7. It selects a random back-off timer to avoid colliding with other waiting devices.
8. Once transmitted, the sender requires an explicit Acknowledgement (ACK) frame. If no ACK arrives, a collision is assumed, the contention window doubles, and the frame is retransmitted.
9. RTS/CTS (Request to Send/Clear to Send) frames can be optionally used to explicitly reserve the medium and solve the hidden node issue.

Key Differences: While CSMA/CD reacts to and recovers from collisions on a wired medium, CSMA/CA actively attempts to prevent them on a wireless medium. CA introduces higher overhead (due to ACKs, timers, and RTS/CTS) compared to the relatively low overhead of CD.

Q8. WLAN Design for a Laboratory (12 m × 8 m)

To reliably support roughly 60 simultaneous devices (workstations and laptops) in a 96 square meter room, a robust wireless architecture is required.

- **Hardware Deployment:** Three ceiling-mounted dual-band Access Points (Wi-Fi 5 or Wi-Fi 6) spaced evenly across the length of the room. This ensures overlapping coverage without dead zones and limits client density to ~20 devices per AP.
- **Frequency & Channel Plan:** 2.4 GHz: Assign non-overlapping channels (1, 6, and 11) to the three APs to eliminate co-channel interference. 5 GHz: Utilize distinct 20 MHz or 40 MHz channels (e.g., 36, 40, 44). Enable band-steering to push capable devices to the 5 GHz spectrum.
- **Infrastructure:** The APs will connect back to a Gigabit PoE (Power over Ethernet) switch, eliminating the need for localized power adapters.
- **Network Addressing:** Implement segmented networks. For example, the Lab network on 192.168.10.0/24 and an isolated Guest network on 192.168.20.0/24. The APs themselves should reside on a static management VLAN.
- **Security Policies:** Enforce WPA3 (or WPA2-Enterprise via 802.1X) for the main lab network. Implement client isolation on the guest SSID.

Q9. The Necessity and Function of Mobile IP

In standard routing, an IP address acts as both a device identifier and its topological location. If a device moves to a new subnet, its IP address must change to route traffic correctly, instantly severing active connections (like TCP sessions). Mobile IP (IPv4 RFC 5944) resolves this by allowing a device to roam across networks while maintaining a constant IP identifier.

Core Mechanisms:

- **Components:** The Mobile Node (MN) roams. The Home Agent (HA) is a router on the permanent network. The Foreign Agent (FA) is a router on the visited network providing a temporary Care-of Address.
- **Operation:** When the MN moves, it registers its new Care-of Address with its HA.
- **Tunnelling:** Any packets sent to the MN's original Home Address are intercepted by the HA. The HA encapsulates these packets (IP-in-IP tunnelling) and forwards them to the FA, which decapsulates and delivers them to the MN.

Mobile IP is crucial for continuous connectivity on devices like smartphones and laptops transitioning between Wi-Fi and cellular networks, though it can introduce inefficiencies like "triangle routing."

Q10. Comparing Ethernet and 802.11 Frames

Because the wireless medium is unguided, shared, and prone to interference, 802.11 frames are inherently more complex than standard wired Ethernet frames.

Feature	Ethernet II (802.3)	Wireless (802.11)
Header Size	Minimal: 14 bytes.	Substantial: 24 to 32 bytes (depends on QoS/Addr4).
MAC Addresses	2 (Source and Destination).	Up to 4 (Transmitter, Receiver, Source, Destination).
Frame Varieties	Strictly Data frames at Layer 2.	Diverse: Data, Management (Beacons), Control (ACKs).
Control Fields	Limited primarily to EtherType.	Extensive: Frame Control (subtypes), NAV Duration, Power Management, Security flags.
Reliability	No Layer 2 Acknowledgements.	Explicit MAC-layer ACKs required for unicast frames.
Security	None built into the standard Layer 2 frame.	Encrypted payloads (WPA2/WPA3) denoted by protected flags.

Conclusion: Wired switches provide dedicated, reliable paths, allowing Ethernet frames to remain simple and lean. Conversely, the chaotic nature of the wireless spectrum necessitates 802.11 frames to carry extensive metadata for collision avoidance (durations), fragmentation (sequence numbers), and error recovery (ACKs).